

An explicit quartic counterexample to Zhao's Vanishing Conjecture

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Abstract

Starting from the three-dimensional polynomial map posted by Levent Alpöge on 20 July 2026, crediting Akhil Mathew for posing the question and Claude Fable for producing the example, we explicitly carry out the Bass–Connell–Wright degree reduction and the de Bondt–van den Essen symmetric reduction. A factor-reusing version of the degree-reduction step gives a cubic homogeneous Keller map in 27 variables, rather than the much larger map obtained by reducing expanded monomials independently. Symmetrization then gives an explicit homogeneous quartic Hessian-nilpotent polynomial in 54 variables, with 598 monomials, whose gradient Keller map is noninjective. Consequently this polynomial is a counterexample to Zhao's Vanishing Conjecture. The expanded polynomial and a two-point collision are supplied as machine-readable exact certificates.

Verification status. The algebra has passed the supplied exact symbolic checks, but this short note has not been peer reviewed. Independent expert scrutiny is welcome.

1 Statement of the result

Let $i^2 = -1$. The straight-line construction below defines a cubic homogeneous map $h : \mathbb{C}^{27} \rightarrow \mathbb{C}^{27}$. For $A, B \in \mathbb{C}^{27}$, set

$$\mathcal{P}(A, B) = i \sum_{j=1}^{27} h_j(A + iB) B_j. \quad (1)$$

Theorem 1. *The polynomial $\mathcal{P}$ is homogeneous of degree four, has 598 monomials when expanded, and its Hessian is nilpotent. The polynomial map*

$$\Gamma : \mathbb{C}^{54} \longrightarrow \mathbb{C}^{54}, \quad \Gamma(Z) = Z - \nabla \mathcal{P}(Z), \quad (2)$$

has Jacobian determinant one and is noninjective. Moreover,

$$\Delta^m \mathcal{P}^m = 0 \quad \text{for every } m \geq 1, \quad \Delta^m \mathcal{P}^{m+1} \neq 0 \quad \text{for infinitely many } m \geq 0. \quad (3)$$

Thus $\mathcal{P}$ is a counterexample to Zhao's Vanishing Conjecture.

The formulas below and ordinary differentiation form a finite exact certificate. The accompanying `potential_sparse.json` is the fully expanded polynomial over $\mathbb{Q}(i)$, and `collision.json` contains two distinct points with the same image under (2).

2 A 13-variable cubic stable model

Write $u = 1 + xy$. Postcomposing the map announced in [4] by the inverse of its linear part gives the identity-linear Keller map $\Phi = (\Phi_1, \Phi_2, \Phi_3)$:

$$\begin{aligned}\Phi_1 &= x - \frac{3}{2}x^2y - \frac{1}{2}x^3z, \\ \Phi_2 &= y + 3xu^2z + 3xy^2(4 + 3xy), \\ \Phi_3 &= u^3z + y^2u(4 + 3xy).\end{aligned}\tag{4}$$

Thus $\det J\Phi = 1$, and the three points

$$(0, 0, -\frac{1}{4}), \quad (1, -\frac{3}{2}, \frac{13}{2}), \quad (-1, \frac{3}{2}, \frac{13}{2})\tag{5}$$

have the same image $(0, 0, -\frac{1}{4})$.

We use the following stable degree-reduction gadget. Suppose a term cPQ is to be removed from the k th coordinate of a current map M . With new variables a, b , replace

$$M_k \mapsto M_k - c(a + P)(b + Q), \quad (a, b) \mapsto (a + P, b + Q).\tag{6}$$

This is a pre- and post-composition of $M \times \text{id}_{\mathbb{A}^2}$ by triangular automorphisms. A previously created output factor may be reused: with one new variable a , one may replace

$$M_k \mapsto M_k - c(a + P)M_\ell, \quad a \mapsto a + P.\tag{7}$$

Both operations preserve the Jacobian determinant and invertibility. A collision $p \neq q$ lifts through (6) by appending $(-P(p), -Q(p))$ and $(-P(q), -Q(q))$, and similarly for (7).

Starting from (4), perform the following operations in order. The symbols a_j, b_j denote the input variables introduced at each step.

step	target	new or reused factors	c
1	Φ_1	$P = x^2, Q = xz + 3y$; add a_1, b_1	$-1/2$
2	Φ_2	$P = 3x^2y, Q = 2z + xyz + 3y^2$; add a_2, b_2	1
3	Φ_2	$P = xy, Q = a_2z + 3xb_2$; add a_3, b_3	-1
4	Φ_3	$P = xy^2, Q = 7y + 3xz + 3xy^2 + x^2yz$; add a_4, b_4	1
5	Φ_3	add $a_5 + a_4xy$; reuse output $b_1 + xz + 3y$	-1
6	Φ_3	reuse output $a_3 + xy$; add $b_6 + a_4b_1 - yb_4$	1

Finally postcompose by the shear

$$M_{b_4} \mapsto M_{b_4} - M_{a_3}M_{b_1}.\tag{8}$$

The resulting map M has 13 variables

$$X = (x, y, z, a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4, a_5, b_6)\tag{9}$$

and degree three. Its linear part L is the identity except for

$$M_{b_1}^{(1)} = b_1 + 3y, \quad M_{b_2}^{(1)} = b_2 + 2z, \quad M_{b_4}^{(1)} = b_4 + 7y.\tag{10}$$

In particular $\det L = 1$. Put

$$\Psi = L^{-1}M = X + H_2(X) + H_3(X),\tag{11}$$

where H_d is homogeneous of degree d . Then $\deg \Psi = 3$, $\det J\Psi = 1$, and Ψ has the following two preimages of the same point:

$$\begin{aligned} p_0 &= (0, 0, -\frac{1}{4}, 0, 0, 0, \frac{1}{2}, 0, 0, 0, 0, 0, 0), \\ p_1 &= (1, -\frac{3}{2}, \frac{13}{2}, -1, -2, \frac{9}{2}, -10, \frac{3}{2}, \frac{3}{4}, -\frac{9}{4}, -6, -\frac{27}{8}, \frac{9}{2}), \\ \Psi(p_0) &= \Psi(p_1) = (0, 0, -\frac{1}{4}, 0, 0, 0, \frac{1}{2}, 0, 0, 0, 0, 0, 0). \end{aligned} \quad (12)$$

Equations (4) and (6)–(11) are a straight-line specification of all components of Ψ ; no omitted choice is involved.

3 The 27-variable cubic homogeneous map

For $X, Y \in \mathbb{C}^{13}$ and $t \in \mathbb{C}$, define

$$h(X, Y, t) = (tH_2(X) + t^2Y, -H_3(X), 0). \quad (13)$$

Every component of h is cubic homogeneous. Let $\mathcal{B}(W) = W + h(W)$. At $t = 1$,

$$\mathcal{B}(X, Y, 1) = (X + H_2(X) + Y, Y - H_3(X), 1). \quad (14)$$

Consequently, if

$$r_j = (p_j, H_3(p_j), 1), \quad (15)$$

then $\mathcal{B}(r_0) = \mathcal{B}(r_1) = (p_0, 0, 1)$. Here $H_3(p_0) = 0$, while

$$H_3(p_1) = (-\frac{17}{4}, -\frac{45}{8}, \frac{99}{16}, 0, \frac{135}{8}, -\frac{9}{2}, -\frac{177}{8}, 0, 0, \frac{9}{4}, \frac{213}{8}, \frac{27}{8}, 0). \quad (16)$$

Lemma 2. *The polynomial matrix Jh is nilpotent.*

Proof. Homogeneity and (11) give

$$\det(I + tJH_2(X) + t^2JH_3(X)) = \det J\Psi(tX) = 1. \quad (17)$$

The Jacobian determinant of $W \mapsto W + h(W)$ is the same determinant by the block determinant formula. Hence $\det(I + Jh(W)) = 1$. Since h is cubic homogeneous, $Jh(sW) = s^2Jh(W)$. Therefore $\det(I + \lambda Jh) = 1$ identically in λ , which is equivalent to nilpotence of Jh . $\square$

4 Symmetrization and the collision

For a polynomial map $h : \mathbb{C}^r \rightarrow \mathbb{C}^r$, de Bondt and van den Essen [2] associate

$$f_h(A, B) = -i \sum_{j=1}^r h_j(A + iB)B_j. \quad (18)$$

Their characteristic-polynomial identity implies that $\text{Hess}(f_h)$ is nilpotent if and only if Jh is nilpotent. Our polynomial in (1) is $\mathcal{P} = -f_h$, so Lemma 2 proves that $\mathcal{P}$ is Hessian nilpotent. It follows immediately that $\det J\Gamma = 1$.

For completeness, the collision can be seen directly. With the linear map $S(x, y) = (x - iy, y)$, differentiation gives

$$S^{-1}\Gamma S(x, y) = (x + h(x), (I + Jh(x)^T)y - ih(x)). \quad (19)$$

Take r_0, r_1 from (15), let $K_j = I + Jh(r_j)^T$, put

$$y_1 = 0, \quad y_0 = K_0^{-1}(-ih(r_1) + ih(r_0)), \quad (20)$$

and set $z_j = S(r_j, y_j)$. Equation (19) and $r_0 + h(r_0) = r_1 + h(r_1)$ show $\Gamma(z_0) = \Gamma(z_1)$. The two 54-tuples are recorded explicitly in `collision.json`; all coordinates have height at most 261. Direct differentiation of the 598-term expansion checks this equality exactly over $\mathbb{Q}(i)$.

5 Failure of the Vanishing Conjecture

Let $\Delta = \sum_j \partial^2 / \partial Z_j^2$. Zhao proved [3] that Hessian nilpotence of a homogeneous polynomial P is equivalent to

$$\Delta^m P^m = 0 \quad \text{for every } m \geq 1. \quad (21)$$

Lemma 2 and the symmetric reduction therefore show that $\mathcal{P}$ satisfies the hypothesis of the Vanishing Conjecture. Zhao's inversion formula writes the formal inverse of $Z - t\nabla P$ as $Z + t\nabla Q_t$, where

$$Q_t = \sum_{m=0}^{\infty} \frac{t^m}{2^m m! (m+1)!} \Delta^m P^{m+1}. \quad (22)$$

If $\Delta^m \mathcal{P}^{m+1} = 0$ for all sufficiently large m , then Q_t is a polynomial. Specializing at $t = 1$ would give a polynomial inverse of Γ , contradicting the explicit collision above. Thus $\Delta^m \mathcal{P}^{m+1} \neq 0$ for infinitely many m , proving Theorem 1.

6 Reproducibility

The repository contains two complementary exact checks. The script `verify.py` rebuilds the construction symbolically and verifies the original map, all six stable reduction gadgets, the cubic homogeneous map, Hessian nilpotence through the structural reduction, and the final collision. The dependency-free script `verify_exported_stdlib.py` uses only Python's standard library and the two JSON files to verify the expanded polynomial's degree and term count and the exact collision. The latter is not a stand-alone proof of Hessian nilpotence. All arithmetic in both checks is exact. The certificate hashes are

```
potential_sparse.json:
556683b30fd9e7b9ecfb4c4d5395ee3be6a8dc5f918da08ed60db8c125f0df28

collision.json:
cbea49f88cc59ba7955a4dbd710c68224590ae310a320369c49f8c63e1f78cae
```

The full source and certificates are available in the public project repository.

7 Scope and novelty

Zhang’s consequence note of 20 July 2026 observes, by Zhao’s equivalence, that the announced three-dimensional counterexample makes the Vanishing Conjecture false in some finite dimension; it explicitly leaves the consequence existential [5]. An earlier public repository [6] gives a 79-variable cubic homogeneous noninjective Keller map. Applying the standard symmetric reduction to that map would produce a 158-variable quartic in principle, but the repository did not supply an expanded quartic witness or its transported collision at the time of our search.

The narrower contribution here is an executed 54-variable, 598-term quartic Hessian-nilpotent polynomial, an exact collision for its gradient map, and a factor-reusing reduction that keeps the intermediate cubic homogeneous map in 27 variables. To our knowledge, after literature and public-code searches on 21 July 2026, no earlier source gives this compact explicit certificate. We do not claim minimality or unconditional worldwide priority; the literature surrounding the announced counterexample is evolving rapidly.

AI-assistance and verification disclosure

The construction, proof organization, verification programs, website, and typeset drafts were developed with extensive assistance from ChatGPT 5.6 Sol. Alec Kriebel takes responsibility for the submission and for preserving the complete source and exact certificates. The algebraic checks are reproducible, but the note has not been peer reviewed and independent expert scrutiny is welcome.

References

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